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INTERNATIONAL ACADEMIC RESEARCH COMPETITION (IARCO) 2026

RESEARCH PROPOSAL

Title: Planetary and Stellar Constraints Affecting the Potential Habitability of Earth-Sized Exoplanets

Category: Junior Category

Proponent: Adnin Sinwan

Institution: Chattogram Cantonment Public College

Class/Grade/Year: Grade 11

Country: Bangladesh

Date of Submission: September 21, 2026

INTRODUCTION

Exoplanets are planets that orbit stars outside of our solar system. Scientists first confirmed the existence in 1995 using the radial velocity method, which measures periodic shifts in the color spectrum of a parent star caused by a shared center of gravity. At that time, a hot giant roughly half the size of Jupiter was detected by the motion of its parent star. Since then, more than 5,500 exoplanets have been found using advanced telescopes like the Hubble and Kepler space telescopes. Today, scientists continue to search for exoplanets in an effort to find a world with Earth-like conditions that could potentially support life.

When evaluating these worlds, astronomers look for the circumstellar habitable zone—the ring around a star where a rocky world can maintain an atmospheric blanket[1]. Even though the idea of a "second Earth" sounds exciting, a planet cannot be considered habitable simply because it sits inside a habitable zone, as this location alone does not guarantee life [1].

Scientists usually spot Earth-sized planets around small, cool stars called M-dwarfs (red dwarfs), such as Proxima Centauri [2] or the TRAPPIST-1 system [3], because they are conspicuous and easier to detect. However, these planets orbit very close to their host stars, exposing them to intense stellar activity, high-energy irradiation, and extreme gravitational forces [4]. This exposure creates harsh conditions where sustaining an atmosphere becomes a major constraint. Planets are subjected to extreme ultraviolet photoevaporative stripping, coronal mass ejections, and strong stellar wind interactions that can make it impossible for an atmosphere—and life—to survive [5].

There are three primary constraints on a planet despite being in a habitable zone: exposure to violent stellar weather [5], gravitational tidal locking [6], and a lack of planetary shielding [7].

Astronomers often examine and detect these factors separately, which prevents us from determining the relative threat each factor poses.

To address this issue, this study looks at confirmed Earth-sized planets in habitable zones and compares three main sets of constraints: stellar weather, planetary mass and retention properties, and tidal locking dynamics [4, 6, 7].

RESEARCH QUESTIONS

This study aims to find out which planetary, stellar, and gravitational factors create the biggest barriers to habitability for Earth-sized planets located inside the habitable zone. Specifically, this research will answer the following questions:

1. **Stellar Radiation:**
   How much extreme ultraviolet (XUV) radiation and flaring do different types of host stars (M, K, and G dwarfs) produce, and how much atmosphere does this strip away from close-in planets?
2. **Tidal Locking:**
   How many of the known Earth-sized habitable zone planets have become tidally locked due to gravity, and does this permanently freeze or boil away their surface conditions?
3. **Planetary Gravity:**
   How do a planet's own mass, size, and surface gravity help it hold onto its atmosphere against strong stellar winds?
4. **Comparison Across Star Types:**
   Is there a statistically significant difference in habitability threats between planets orbiting cool M-dwarf stars and planets orbiting warmer stars like our Sun?
5. **Ranking the Threats:**
   Which single factor—stellar storms, tidal locking, or low planetary mass—is the strongest constraint on habitability, and which planets should telescopes like the JWST focus on first?

LITERATURE REVIEW

The idea of the habitable zone was first calculated mathematically by researchers who wanted to know how close or far a planet could be from its star to keep liquid water on its surface [8]. Their climate models showed that if a planet is too close, it enters a runaway greenhouse state where its oceans boil off, and if it is too far, carbon dioxide freezes out and turns the world into an iceball [8]. While this concept has been useful for finding candidate planets, newer research shows that distance alone does not tell the whole story [9].

One big problem comes from the type of star the planet orbits. Red dwarfs, also known as M dwarfs are small and dim. Because of this their habitable zones lie close to the star [4].. This closeness comes with serious risks. Young M dwarfs stay extremely active for hundreds of millions of years. They send out bursts of X-ray and ultraviolet (XUV) radiation, which can harm nearby planets [5] [10]. This prolonged high-energy radiation can cause hydrodynamic escape, where the planet's upper atmosphere heats up and boils away into space [11]. Space telescopes like TESS have recorded frequent superflares on these small stars, confirming that planets around M dwarfs face continuous space weather that solar-type stars do not produce [10].

At the same time, strong gravitational pulls create tidal locking [6]. Because habitable zone planets around small stars orbit so close, tidal forces quickly slow down their rotation until one side permanently faces the star while the other stays in darkness [12]. Climate simulations suggest that thick clouds could form over the dayside and help cool the planet down, but this only works if the planet still has enough atmosphere left to move heat around [13]. Recent observations from the James Webb Space Telescope (JWST) have looked at the inner planets of the TRAPPIST-1 system and found little to no thick atmosphere, supporting the concern that close-in worlds struggle to hold onto their air [14].

Finally, a planet's own physical properties dictate whether it can survive these threats. A planet with higher mass and surface gravity has a higher escape velocity, making it harder for stellar winds to strip its air [11]. While many scientists believed that an intrinsic magnetic field would shield an atmosphere, recent space physics models show that magnetic lines can sometimes funnel charged particles directly into the atmosphere and increase gas loss [15].

Overall, existing papers have analyzed solar flares, tidal forces, and planetary mass as separate problems [9]. What is missing is a direct comparison that ranks all these factors together across discovered exoplanets to determine which specific barrier is the most destructive [8], [12]. This research proposal focuses directly on filling that gap [12]

PROPOSED METHODOLOGY

This study uses a quantitative and computational approach to evaluate and rank the physical constraints on exoplanet habitability. The research will be carried out in four structured phases:

1. Data Collection and Target Selection

- Data will be gathered directly from the NASA Exoplanet Archive using its Planetary Systems table.
- Only confirmed exoplanets with a radius equal to or less than 1.5 times the radius of Earth **($R_p \leq 1.5 R_{Earth}$)** will be included. This cutoff ensures that only rocky, terrestrial worlds are studied, filtering out gas-heavy mini-Neptunes.
- The sample will be restricted to planets located inside their host star's habitable zone, calculated based on the incident stellar flux limits from established climate models.
- For each planet, key parameters will be downloaded: planet radius, planet mass, orbital period, semi-major axis, orbital eccentricity, stellar effective temperature, stellar mass, and star age. If a planet has a measured radius but an unmeasured mass, standard mass-radius empirical equations will be used to estimate its mass.

2. Calculating the Constraint Factors

Using Python and spreadsheet analysis, three main constraint values will be calculated for every planet in the dataset:

- **Planetary Escape Velocity (v_esc):**
- To see how well the planet can hold onto its air, surface gravity and escape velocity will be calculated using:
  $v_{esc} = \sqrt{((2 \times G \times M_p) / R_p)}$
  A lower escape velocity means the planet is more likely to lose its atmosphere to space.
- **Atmospheric Stripping Rate:**
  The amount of atmosphere lost over time will be calculated using the standard energy-limited escape formula:
  $M_{loss} = (\eta \times \pi \times R_p^3 \times F_{XUV}) / (G \times M_p)$
  Where **η** is the heating efficiency, and F_{XUV} is the high-energy radiation coming from the star based on its spectral type.
- **Tidal Locking Timescale:**
  The time it takes for the host star's gravity to lock the planet into permanent day and night sides will be calculated using standard tidal equations. If the calculated locking time is shorter than the star's current age, the planet is categorized as tidally locked.

3. Comparing and Ranking the Threats

- To compare factors with different units (like radiation vs. gravity), all values will be standardized on a normalized scale from 0 to 1.
- A composite Habitability Constraint Score will be calculated for each planet by combining its radiation score, tidal score, and retention score.

- The planets will be grouped by their host star types: M dwarfs (red dwarfs), K dwarfs (orange stars), and G dwarfs (yellow, Sun-like stars) to see which star type creates the most hostile conditions.

4. Statistical Analysis

- Basic descriptive statistics—including the mean, median, and standard deviation—will be computed for all constraint scores across each star type.
- A One-Way Analysis of Variance (ANOVA) will be performed to test if there is a statistically significant difference ($p < 0.05$) in atmospheric loss and tidal danger between planets around M dwarfs versus Sun-like stars.
- Finally, the factors will be ranked in order of their impact to identify which single barrier—space radiation, tidal locking, or low planet mass—is the primary obstacle to keeping a habitable surface.

EXPECTED OUTCOME

This study aims to create an data-driven ranking of the physical barriers that could make Earth-sized planets less habitable. The analysis will determine whether energy stellar radiation or gravitational tidal locking is the bigger challenge for holding onto an atmosphere across different types of stars. By identifying which exoplanets have the overall threat levels the final ranking will serve as an objective tool for astronomers deciding which planets to observe closely with powerful telescopes like the James Webb Space Telescope.

POTENTIAL LIMITATIONS

Observational Bias in the Data: The NASA Exoplanet Archive has more planets discovered around small M-dwarf stars compared to Sun-like stars. This imbalance happens because transits around smaller dimmer stars are easier to detect. As a result there are Earth-sized planets orbiting G-type stars, which limits how well the results can represent true Earth analogues.

Simplified Atmospheric Models: The energy-limited escape equation gives an estimate of how quickly gases can escape a planet's atmosphere. It does not include chemistry or real-time magnetic field effects. These missing factors might change retention in some cases though the model still provides a useful first approximation.

CONCLUSION

Finding an Earth- planet in a star's habitable zone is just the beginning. To truly assess if a world could support life we must understand the threats it faces. This study will. Compare the dangers from stellar radiation, tidal locking and weak gravity. The results will help identify which conditions reduce the chance of keeping an atmosphere. Ultimately this work will guide telescope time toward the promising targets—planets that are more likely to keep their air and potentially sustain life.

PROJECT PRACTICALITIES

The project is expected to take eight months.

Months 1–2: Pull data from the NASA Exoplanet Archive clean the datasets. Narrow down the sample, to Earth-sized planets in the habitable zone.

Months 3–4: Develop Python scripts to compute escape velocities, atmospheric loss rates and tidal synchronization timescales.

Months 5–6: Run analyses standardize threat scores and apply ANOVA tests to compare different star types.

Months 7–8: Finalize the ranking of threats write the project report and prepare presentation materials for sharing the results.

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